Definition 9.1

Regression analysis is a mathematical measure of the average relationship between two or more variables in terms of the original units of the data.

9.3.1 Dependent and independent variables

Definition 9.2

In regression analysis there are two types of variables. The variable whose value is to be predicted is called dependent variable and the variable which is used for prediction is called independent variable.

Regression helps us to estimate the value of one variable, provided the value of the other variable is given. The statistical method which helps us to estimate the unknown value of one variable from the known value of the related variable is called Regression.

9.3.2 Regression Equations

Regression equations are algebraic expressions of the regression lines. Since there are two regression lines, there are two regression equations. The regression equation of X on Y is used to describe the variation in the values of X for given changes in Y and the regression equation of Y on X is used to describe the variation in the values of Y for given changes in X. Regression equations of (i) X on Y (ii) Y on X and their coefficients in different cases are described as follows.

Case 1: When the actual values are taken

When we deal with actual values of X and Y variables the two regression equations and their respective coefficients are written as follows:

(i) Regression Equation of X on Y:

$$X - \overline{X} = b_{xy}(Y - \overline{Y})$$

where $\overline{X}$ is the mean of X series,

$\overline{Y}$ is the mean of Y series,

$b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{N\Sigma Y^2 - (\Sigma Y)^2}$ is known as the regression coefficient of X on Y, and r is the correlation coefficient between X and Y, σ_x and σ_y are standard deviations of X and Y respectively.

(ii) Regression Equation of *Y* on *X*;

$$Y - \overline{Y} = b_{yx}(X - \overline{X})$$

where $\overline{X}$ is the mean of X series,

$\overline{Y}$ is the mean of Y series,

$b_{yx} = r\frac{\sigma y}{\sigma x} = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{N\Sigma Y^2 - (\Sigma Y)^2}$ is known as the regression coefficient of Y on X, and r is the correlation coefficient between X and Y, σ_x and σ_y are standard deviations of X and Y respectively.

Case 2: Deviations taken from Arithmetic means of X and Y

The calculation can very much be simplified instead of dealing with the actual values of X and Y, we take the deviations of X and Y series from their respective means. In such a case the two regression equations and their respective coefficients are written as follows:

(i) Regression Equation of X on Y:

$$X - \overline{X} = b_{xy}(Y - \overline{Y})$$

where $\overline{X}$ is the mean of X series,

$\overline{Y}$ is the mean of Y series,

$b_{xy} = r\frac{\sigma x}{\sigma y} = \frac{\Sigma xy}{\Sigma y^2}$ is known as the regression coefficient of X on Y, $x = (X - \overline{X})$ and $y = (Y - \overline{Y})$

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